

Architecture

Contents

Contents	2
md2tex	3
Core Modules	3
Flow of Data	3

md2tex

md2tex is built around a layered architecture cleanly separating text parsing, abstract representations, escaping mechanisms, and output rendering.

Core Modules

- **cli.py & tui/**: User interface layers routing interactions via Click (CLI commands) or Textual (Terminal User Interface). Both interface layers utilize the `md2tex/services.py` core to execute Markdown conversions reliably.
- **services.py**: The orchestration layer. It receives requests, standardizes path interactions, constructs `RenderOptions`, calls the parser and renderer, and injects templates.
- **parser.py**: A block-level, line-by-line Markdown parser. It scans strings and safely extracts raw block entities (like `CodeBlock`, `BlockQuote`, `Table`, `Heading`) into AST IR Nodes.
- **nodes.py**: Simple structure layer outlining the abstract tree (`Table`, `Heading`, `CodeBlock`, `FootnoteDef`, `ListBlock`, etc.).
- **escaping.py**: Security translation module correctly escaping character artifacts (`$`, `%`, `{}`) *after* blocks are parsed, but *before* formatting runs.
- **renderer.py**: Main TeX compilation module processing IR Nodes. It handles each node type (e.g., `_render_table`) and recursively processes Inline Formatting via Regex (Italics, Hyperlinks, Strikethrough, etc.).

Flow of Data

Markdown String -> Block Parser -> IR Ast (`nodes.py`) -> Renderer (escape -> format -> inject) -> TeX String